An initial Θ-datum over an imaginary quadratic field, and numerical values of the local quantities of inter-universal Teichmüller theory IV

A second report

Toshihisa Urahigashi
123026.urahigashi.toshihisa@gmail.com

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ORCID 0009-0004-0460-6242

Abstract

We exhibit an initial Θ-datum in the sense of [IUT1, Def. 3.1] whose field of moduli is the *imaginary quadratic* field $\mathbb{Q}(i)$: the curve $E_0 : y^2 = x(x-1)(x-a)$ with $a = (10+3i)^{29}$ and $\ell = 7$, for which the rational prime 109 splits, so that one place of multiplicative and one place of good reduction lie above it. All conditions (a)–(f) are verified; the arithmetic is exact and reproduced by a short script, and the geometric conditions (d)–(f) are established in §4. The local invariants are $e_{\mathrm{g}} = 1$, $e_{\mathrm{b}} = 105$ and $\mathrm{ord}_{109}(\underline{\underline{q}}_{\mathrm{b}}) = 29/7$, so that the *ramification* invariants of this datum agree with those of the datum over $\mathbb{Q}(\sqrt{5})$ in [First]. The residual data do not: the residue degrees are $f_{\mathrm{g}} = 12$, $f_{\mathrm{b}} = 6$ here against 24 and 2 there, and the mixed tensor algebra has 6 components of degree 1260 against 2 of degree 2520 *(the values at* 109 *are computed in* `scripts/tables.py`*, Table 13, from the point count and the order of Frobenius on* $E_0[3]$, $E_0[5]$, $E_0[7]$*; the values at* 211 *are those of [First]. The consistency check* $12 \cdot 630 = 7560 = 6 \cdot 1260$ *is asserted there.)* Everything tabulated in §15 is computed at 211 and does not transfer.

Explicit initial Θ-data are needed for any numerically effective use of the theory. Dupuy and Hilado [DH25] construct such data systematically for elliptic curves over $\mathbb{Q}$, working out the curve of Cremona label 11a1 in full and reporting 5525 curves of prime conductor for which some ℓ works. The datum below is over $\mathbb{Q}(i)$; the point of it is not that an explicit datum is rare but that the *configuration* it carries is unavailable over $\mathbb{Q}$. It is also, in three specific respects, easier to construct than the datum of [First]: the nodes are already split over the field of moduli, the degree bound is halved, and the existence family needs two congruences instead of five, with the condition on ℓ becoming automatic (§6). We give that family as Proposition 2: for every N there are infinitely many *arithmetic parameters* of this shape over $\mathbb{Q}(i)$ with $\mathrm{ord}_\pi(a) = N$, so the configuration is not isolated. *Turning one of them into an initial* Θ*-datum needs more*: $7 \nmid N$ (else $\ell = 7$ divides the Tate order $2N$), the Galois image condition, and the geometric choices of §4; the proposition is an arithmetic statement and does not assert them.

The effective results of [MFHMP] are stated for a mono-complex field, that is, for $\mathbb{Q}$ or an imaginary quadratic field [MFHMP, Thm. A], with an explicit constant in the degree-two case, and the curve there is in the same Legendre form; the datum of [First] lies over a real quadratic field, and the one given here does not. *Whether that bears on the estimate is not addressed here.*

Explicit data also make the quantities in the local estimates of [IUT4, Thm. 1.10] numerical at these inputs. *Explicit formulas, and the plan of evaluating them, are already in [IUT4],* *[MFHMP] and [DHPS]*; what is added here is their exact evaluation on the two data and a set of checks that detects the kind of error §9 records. Sections 8–17 tabulate them for this datum and for that of [First]. Every entry is either a formula printed in the source, marked [S], or a computation from it, marked [P]; each table names the locator of the quantity it tabulates, and every number is reproduced by `scripts/tables.py`, which uses only the standard library — with one exception, named where it occurs: the counts of the Cremona sweep in §9 are transcribed from an external computation, and $\mathrm{ord}_3(q) = 44$ at the two curves it returns is taken from that computation as an input, the arithmetic that follows from 44 being what is recomputed. Three of the tables evaluate, on these explicit inputs, three questions: where the data on record sit relative to the threshold at which the stage-one comparison ceases to be vacuous (§9); which region of the parameters (ℓ, p, N) admits a datum satisfying at once that threshold, the choice of ℓ made in [IUT4, Cor. 2.2(ii)], and the condition $I^* = \varnothing$ of [IUT4, Prop. 1.4(iii)] (§12); and how the two terms of the averaged upper bound compare with each other at these parameters (§17). For a candidate that already satisfies the other three, that region is a window $\sqrt{H_\forall} \leq \ell < N/3$, its left wall the lower half of the source's condition (C1) and its right wall the threshold derived here — the *upper* half of (C1) is checked separately and is not binding on any datum tabulated here — and §13 exhibits a single explicit curve over $\mathbb{Q}(i)$ whose window contains *fourteen* primes, with two distinct upper constraints simultaneously tight at $\ell = 157$. *Scope*: this report constructs data and evaluates local quantities on them. It does not compute the hull of the union of the possible images of the actual Θ-pilot or compare the two sides of [IUT3, Cor. 3.12]; nothing here is conditional on the hypothesis (SA) of [First]; and no claim is made about [IUT3, Cor. 3.12], about [MFHMP, Thm. A], about the objections of [SS], or about *abc*.

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1 What this is, and what it is for

[P] An initial Θ-datum is a list of conditions [IUT1, Def. 3.1] that a curve, a prime ℓ, a set of bad places, a section and a cusp must jointly satisfy. Verifying the list against an explicit curve is a finite task, and doing it produces an object that others can compute with. [S] [DH25] does this over $\mathbb{Q}$: it breaks [IUT1, Def. 3.1] into pieces, works out the curve 11a1 in full, and reports that for 5525 curves of prime conductor in the Cremona tables some $\ell \in \{7, 11, 13\}$ makes the mod-ℓ representation over $\mathbb{Q}(i, E[30])$ surjective — that one condition, not the whole of [IUT1, Def. 3.1]. [P] *Over $\mathbb{Q}$ the mixed profile cannot appear at all*: by [First, Cor. 3(i)] every place tuple above every prime is constant when $F_{\mathrm{mod}} = \mathbb{Q}$, so no member of that family carries it, and the field must be at least quadratic for the configuration to exist. [First] gives a datum over $\mathbb{Q}(\sqrt{5})$ that does carry it; the present report gives one over $\mathbb{Q}(i)$, and shows that the construction is in several respects easier there (§6). The value of this does not depend on any reading of the theory, on any conclusion about *abc*, or on the outcome of any dispute: a verified datum is a verified datum.

[P] Two further things come out of the construction. First, the configuration in which a rational prime splits in the field of moduli and carries one place of multiplicative and one of good reduction occurs here, as it does in [First]; by [First, Cor. 3(i)] it cannot occur when $F_{\mathrm{mod}} = \mathbb{Q}$, so the field must be at least quadratic for it to appear at all. Second, it is not isolated: Proposition 2 produces infinitely many arithmetic parameters of this shape over $\mathbb{Q}(i)$, of arbitrary depth at the split prime — subject, for the passage to a datum, to $7 \nmid N$ and to the conditions of §4.

Remark 1 (where such data are used)**.** [S] [MFHMP, Thm. A] is stated for "a mono-complex number field [i.e., $\mathbb{Q}$ or an imaginary quadratic field — cf. Definition 1.2]", Definition 1.2 there reading "F is mono-complex if F is either the field of rational numbers $\mathbb{Q}$ or an imaginary quadratic field". It provides the constant

$$h_d(\epsilon) = \begin{cases} 3.4 \cdot 10^{30}\, \epsilon^{-166/81} & (d=1), \\ 6 \cdot 10^{31}\, \epsilon^{-174/85} & (d=2), \end{cases} \qquad d = [L:\mathbb{Q}],$$

so the degree-two case is provided for explicitly; and that the curve there is $E_{a,b,c} : y^2 = x(x-1)(x+a/c)$, the Legendre form used here. The datum of [First] lies over a real quadratic field and so is not of that kind; the datum below is. *That is a statement about where the two data lie, and nothing more: whether the configuration bears on the estimate of [MFHMP, Thm. A] is an open question which this report does not address and on which none of its claims depend.*

Remark 2 (independence)**.** [P] [SS], [Rpt] and [LANA] are listed because a reader will want to know where this sits relative to them; *no statement below uses any of them, and no claim below is conditional on how the questions they are about are answered.* In particular nothing here depends on the hypothesis (SA) of [First]. What is not computed is the hull of the union of the possible images of the actual Θ-pilot, and what is not compared is the two sides of [IUT3, Cor. 3.12]: what is asserted is a construction and its verification.

2 The datum

$$\begin{aligned} &L = F_{\mathrm{mod}} = \mathbb{Q}(i), && \pi = 10+3i, \quad \mathrm{N}(\pi) = 109, \quad a = \pi^{29}, \\ &E_0 : y^2 = x(x-1)(x-a), && \ell = 7, \\ &F = L(E_0[15]), && K = F(E_0[7]) = L(E_0[105]). \end{aligned} \tag{1}$$

Let $X_F = E_F \setminus \{0\}$. Select as $\underline{\mathbb{V}}^{\mathrm{bad}}_{\mathrm{mod}}$ all places of L of multiplicative reduction outside residue characteristics 2 and 7, including the place v_π.

Theorem 1. [P] (1) *extends to an initial* Θ*-datum in the sense of [IUT1, Def. 3.1]. Above* 109 *the set* $\mathbb{V}$ *has exactly two elements, one selected bad and one good, and*

$$e_{\mathrm{g}} = 1, \qquad e_{\mathrm{b}} = 105, \qquad \mathrm{ord}_{109}(\underline{\underline{q}}_{\mathrm{b}}) = \frac{29}{7},$$

so every local index in this profile satisfies $e \le p-2 = 107$*.*

The arithmetic conditions are verified in §3 and the geometric ones in §4. *The slack* $107-105 = 2$ *is small, but the inequality* $105 \le 107$ *is what is required; §7 lists roomier primes.*

3 The arithmetic conditions

The finite integer, rational and residue-field computations of this section are exact and are reproduced by `scripts/imq_checks.py`. *Two things in this section are not*: the seventh-power-free sieve, which is a separate program over all primes below its bound (§20), and the appeals to cited structure theorems, which are applications and not computations.

(a) The field of moduli. Writing $\bar{a}$ for the conjugate of a, the six Legendre orbit comparisons $\bar{a} \in \{a, a^{-1}, 1-a, (1-a)^{-1}, a(a-1)^{-1}, (a-1)a^{-1}\}$ all fail in $\mathbb{Z}[i]$. Hence $j(E_0) \ne \overline{j(E_0)}$ and $F_{\mathrm{mod}} = \mathbb{Q}(j(E_0)) = \mathbb{Q}(i)$. Computing j exactly in $\mathbb{Q}(i)$, its imaginary part, in lowest terms, is a nonzero integer of 296 digits over a denominator of 234 digits.

The mixed profile above 109. Since $109 = \pi\bar{\pi}$ splits and $a = \pi^{29}$,

$$\mathrm{ord}_\pi(a) = 29, \qquad \mathrm{ord}_{\bar{\pi}}(a) = 0, \qquad \mathrm{ord}_\pi(1-a) = \mathrm{ord}_{\bar{\pi}}(1-a) = 0.$$

So E_0 has multiplicative reduction at π and good reduction at $\bar{\pi}$, and $\mathbb{V}$ has one selected bad and one good element above 109.

(b) Selected places, and the extra hypothesis of [IUT4]. $\mathrm{N}(a) = 109^{29}$ and

$$\mathrm{N}(1-a) = 2 \cdot 3^2 \cdot 5 \cdot 21577 \cdot C,$$

with C of 53 digits. The full norm is seventh-power-free: the exhaustive sieve of `scripts/imq_sieve7.cpp` tests every prime up to $\lfloor C^{1/7} \rfloor = 34{,}869{,}670$ and finds none. Primality of C is not asserted. Every selected place has odd residue characteristic, and $7 \nmid \mathrm{N}(1-a)$. Since the bad set consists of *all* multiplicative places outside 2 and 7, the additional good-reduction hypothesis of [IUT4, Thm. 1.10] holds by construction.

(b) Stable reduction. [S] [IUT1, Def. 3.1(b)] asks for stable reduction at every $v \in \mathbb{V}(F)^{\mathrm{non}}$. [IUT4, Prop. 1.8(v)] gives it for a curve all of whose l-torsion is rational, for one prime $l \geq 3$ *invertible in* $\mathcal{O}_k$. [P] Here $E_0[15] \subset E_0(F)$ by construction, so both $l = 3$ and $l = 5$ are available; a nonarchimedean place has a single residue characteristic, so at least one of them is invertible there. *That is what the choice of* $15 = 3 \cdot 5$ *buys*: neither 3 nor 5 alone would cover its own residue characteristic. Marking the origin and contracting the components that this destabilises carries the semistable model of E_0 to a stable model of the marked curve of type $(1,1)$, which is what [IUT1, Def. 3.1(b)] asks for. No further condition on a is needed.

(c) The order at every selected place, and the mod-ℓ image. By the norm identity $v_q(\mathrm{N}(z)) = \sum_{\mathfrak{p}|q} f(\mathfrak{p}/q) v_{\mathfrak{p}}(z)$, seventh-power-freeness gives $v_{\mathfrak{p}}(1-a) \leq 6$ at every prime ideal, hence prime to 7. At v_π the base Tate order is $2 \cdot 29 = 58$ and $7 \nmid 58$.

For the image: at v_π the Tate inertia contains the transvection $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, nontrivial because $7 \nmid 58$. Each rational prime $q \equiv 1 \pmod 4$ has *two* places of $\mathbb{Q}(i)$ above it, and they *may* give different Frobenius traces, so a witness is a place and not merely a prime. (They need not: at $q = 233$ both places send a to 12, so both give trace -26.) Take the place at which i reduces to the smaller root r of $r^2 \equiv -1$:

q	13	17	37	41	53	61
r	5	4	6	9	23	11
a_q	6	-6	-2	-6	-6	-10
$a_q^2 - 4q \bmod 7$	5	3	3	5	6	3

Each discriminant is a nonsquare modulo 7, so $X^2 - a_q X + q$ is irreducible there. *The choice of place is not cosmetic*: at $q = 13$ the other place has $a_q = 2$ and discriminant 1, and at $q = 61$ it has $a_q = -6$ and discriminant 2, both squares, so those two places are not witnesses. [P] *The passage from these facts to the image is made without choosing a basis.* Write L for the fixed line of T and g for a Frobenius at one of the six places. Since the characteristic polynomial of g is irreducible, g has no eigenvector, so $gL \neq L$; hence T and gTg^{-1} are nontrivial transvections with *distinct* fixed lines. As $\mathrm{SL}_2(\mathbb{F}_7)$ is transitive on ordered pairs of distinct lines, the pair may be moved to $(\langle e_1 \rangle, \langle e_2 \rangle)$, where the two transvections read $\left(\begin{smallmatrix} 1 & a \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 0 \\ b & 1 \end{smallmatrix}\right)$ with $a, b \neq 0$; conjugating further by $\mathrm{diag}(a^{-1}, 1)$ makes $a = 1$. Each of the six remaining pairs generates a group of order

$336 = |\mathrm{SL}_2(\mathbb{F}_7)|$. Every step is checked in the kernel in `lean/BasisFreeImage.lean`: that all 882 matrices with irreducible characteristic polynomial move all eight lines, that the orbit of $(\langle e_1\rangle, \langle e_2\rangle)$ has all 56 elements, and the six closures.

Remark 3 (correction)**.** [P] An earlier version conjugated T by the *companion matrix* of the characteristic polynomial and checked that one pair. That does not settle the image: the actual Frobenius is only conjugate to the companion matrix, and the basis putting inertia in the form T need not be the one putting Frobenius in companion form. The argument above quantifies over all matrices with irreducible characteristic polynomial and over all pairs of distinct lines, so no basis is chosen anywhere.

Finally $[F : F_{\mathrm{mod}}]$ divides

$$|\mathrm{GL}_2(\mathbb{Z}/15\mathbb{Z})| = 23040 = 2^9 \cdot 3^2 \cdot 5,$$

which is prime to 7, while $|\mathrm{PSL}_2(\mathbb{F}_7)| = 168 = 2^3 \cdot 3 \cdot 7$ is divisible by 7; every proper normal subgroup of $\mathrm{SL}_2(\mathbb{F}_7)$ lies in its centre. So intersecting with the image over F cannot give a proper subgroup, and $\mathrm{Gal}(K/F)$ has the required image.

The local invariants. At $\bar{\pi}$ the reduction is good and the torsion is prime to 109, so $e_{\mathrm{g}} = 1$. At π the curve is split Tate with $\mathrm{ord}_{109}(q) = 58$; since $\gcd(105, 58) = 1$, adjoining $E_0[105]$ has ramification index exactly 105, whence $e_{\mathrm{b}} = 105$, and $\mathrm{ord}_{109}(\underline{\underline{q}}_{\mathrm{b}}) = 58/14 = 29/7$.

4 The geometric conditions

Split reduction at every selected bad place. a is integral over $\mathbb{Z}[i]$. At an odd place where a and $a-1$ are both units the discriminant $16a^2(a-1)^2$ is a unit, so only the two nodal reductions need inspection. If $a \equiv 0$ the special fibre is $y^2 = x^2(x-1)$ with tangent cone $y^2 = -x^2$, which splits because $i \in L$. If $a \equiv 1$, putting $t = x - 1$ gives $y^2 = (1+t)t^2$ with tangent cone $y^2 = t^2$, already split. The residue characteristics are odd, so the branches are distinct. *Reduction is therefore split multiplicative already over L at every selected bad place, and stays split over F and K.* This is not deduced from rational 2-torsion alone.

(d) Quotient and core. Write $V = E_0[7](K)$, fix a line $W \subset V$ and $0 \neq \xi \in Q = V/W$, and let $\pi_W : E_K \to E' = E_K/W$ with $\psi : E' \to E_K$, $\psi\pi_W = [7]$. Restriction gives $\bar{\pi}_W : Q \simeq \ker\psi$, so this kernel is constant over K. Put $\underline{X}_K = E' \setminus \ker\psi$, $\underline{C}_K = [\underline{X}_K/\{\pm 1\}]$ and $C_K = [X_K/\{\pm 1\}]$. The cover $\underline{X}_K \to X_K$ has geometric quotient Q; its extension to the compactifications is unramified above the zero cusp with K-rational fibre there, so the cusp decomposition group maps trivially to Q. Passing to the inversion quotients gives type $(1, 7\text{-tors})^{\pm}$ as in [EtTh, Def. 2.1]. Let ϵ be the cusp corresponding to $\pm\bar{\pi}_W(\xi)$.

By [MFHMP, Prop. 2.1] an arithmetic once-punctured elliptic curve in characteristic zero has one of the four *rational* invariants $2^{14}31^3 5^{-3}$, $2^2 73^3 3^{-4}$, 1728, 0. The present j is not rational, so $X_{\overline{K}}$ — where $X_K := E_K \setminus \{0\}$ is the once-punctured curve — is nonarithmetic; the finite étale map $X_{\overline{K}} \to C_{\overline{K}}$ puts these orbicurves in the same commensurability class, so $C_{\overline{K}}$ is nonarithmetic as well. It is a punctured hemi-elliptic orbicurve, so [CanLift, Prop. 2.7] makes it a $\overline{K}$-core, and [CanLift, Prop. 2.3(i)] descends this to C_K, which is then the core of the finite étale cover $\underline{C}_K$ required by [IUT1, Def. 3.1(d)]. For a completion K_v, extend $K \hookrightarrow \overline{K}$ to $\overline{K} \hookrightarrow \overline{K_v}$ and apply [CanLift, Prop. 2.3(ii)] to these algebraically closed fields of characteristic zero, then (i) again; base change preserves finite étaleness. This covers the complex completions, and no real completion occurs since $i \in K$. The geometric theta covers of (d) are supplied by [EtTh, Prop. 2.2, Def. 2.3]. *We note that j is not an algebraic integer, so E_0 has no complex multiplication; but no non-CM criterion is used above.*

(e) and (f): places and cusp, simultaneously. At each selected bad moduli place u choose a place of F above u and a place w of K above that. The split Tate sequence $0 \to \mu_7 \to E(K_w)[7] \to \mathbb{Z}/7 \to 0$, with the class of $q^{1/7}$ mapping to 1, determines a marked flag $(W_w, \pm\xi_w)$ in V: changing the root moves the lift only by μ_7, and reversing the graph orientation changes the sign. The group $\mathrm{SL}_2(V)$ acts transitively on such flags — lift a nonzero quotient class to t and choose $s \in W$ with $\det(s,t) = 1$ — so, by the image statement of §3, there is $\sigma_u \in \mathrm{Gal}(K/F)$ with $\sigma_u(W_w, \pm\xi_w) = (W, \pm\xi)$. Put $v(u) = \sigma_u w$, with the convention $|b|_{\sigma_u w} = |\sigma_u^{-1} b|_w$; the resulting isomorphism $\sigma_u^{-1} : K_{v(u)} \simeq K_w$ commutes with localization, so the fixed global pair acquires the required local marking at $v(u)$. *The global W and ϵ are not replaced place by place.*

Since σ_u fixes F, the place $v(u)$ lies above the same u; distinct moduli-place fibres are disjoint, so *the choices at different u do not interfere and can be made simultaneously.* Choosing one arbitrary lift at each remaining moduli place gives $\mathbb{V} \xrightarrow{\sim} \mathbb{V}_{\mathrm{mod}}$ as required by [IUT1, Def. 3.1(e)], which asks for a set-theoretic section and imposes neither equivariance nor a single common Galois element.

Locally the quotient is $E(q) \to E(q^7)$, $z \mapsto z^7$, with dual $E(q^7) \to E(q)$, $z \mapsto z$; the latter multiplies the period of the graph by 7. *Two things have to hold, and we separate them so that each can be checked against [EtTh, Def. 2.5]. First*, the fundamental-group quotient factors through $\widehat{\mathbb{Z}} \to \mathbb{Z}/7$, which is the displayed local picture. *Second*, the chosen cusp splitting is compatible with the $\pm$ structure: since $\ker\psi = \{q^r \bmod q^{7\mathbb{Z}} : r \in \mathbb{Z}/7\}$ and $\pi_W(q^{1/7}) = q$, the localization of the fixed cusp ϵ *is* the canonical ± 1 cusp of [IUT1, Def. 3.1(f)]: the label counts deck translations by the period q, and no Tate order or ramification index enters it. Rational full 2-torsion gives the field condition of [IUT1, Def. 3.1(e)]; the Weil pairings give $\mu_{12} \subset F$ and $\mu_7 \subset K$.

5 An arithmetic existence family

The configuration is not isolated. Put $f(T) = (T^2+1)((T-1)^2+1) = T^4 - 2T^3 + 3T^2 - 2T + 2$, of discriminant $400 = 2^4 5^2$.

Proposition 2. [P] *Let p be a rational prime that splits in $\mathbb{Q}(i)$ with $p \neq 5$ — equivalently here $p \geq 13$ — and let N be a positive integer. There are infinitely many positive integers A such that, for $a = A - i$ and the two places $\pi, \bar\pi$ of $\mathbb{Q}(i)$ above p,*

$$\mathrm{ord}_\pi(a) = N, \qquad \mathrm{ord}_{\bar\pi}(a) = 0, \qquad \mathrm{ord}_\pi(1-a) = \mathrm{ord}_{\bar\pi}(1-a) = 0,$$

and $\mathrm{N}(a)\,\mathrm{N}(1-a)/p^N$ is a positive seventh-power-free integer.

Construction. Hensel gives r_N with $r_N^2 \equiv -1 \pmod{p^{N+1}}$ and $r_N \equiv r_1 \pmod p$. Impose

$$A \equiv r_N + p^N \pmod{p^{N+1}}, \qquad A \equiv 0 \text{ or } 1 \pmod 5,$$

put $M = 5p^{N+1}$, $A = A_0 + Mx$ and $G_N(x) = p^{-N} f(A_0 + Mx)$, a quartic in x. *The exclusion of $p = 5$ is needed for the two congruences to be simultaneously solvable*: if $p = 5$ then $r_N \equiv \pm 2 \pmod 5$, so $A \equiv r_N + 5^N \equiv \pm 2 \pmod 5$ for $N \geq 1$, which meets neither $A \equiv 0$ nor $A \equiv 1$; the moduli 5 and p^{N+1} must also be coprime for the two to be combined. The first congruence forces

$\mathrm{ord}_\pi(a) = N$ exactly; since $\pi \mid (A-i)$ implies $\pi \nmid (A+i)$, this is $v_p(A^2+1)$. *The prime p itself does not divide the other factor*: $A \equiv r_1 \pmod p$, so

$$(A-1)^2 + 1 \equiv (r_1-1)^2 + 1 = 1 - 2r_1 \pmod p,$$

and if this vanished then $2r_1 \equiv 1$, whence $4r_1^2 \equiv 1$ and, using $r_1^2 \equiv -1$, $p \mid 5$ — excluded. Hence $v_p(f(A)) = v_p(A^2+1) = N$ and $p \nmid G_N(x)$.

Only two congruences are needed, because of the following three facts, each checked exhaustively. *(i)* $v_2(f(A)) = 1$ for every A: if A is odd then $A^2+1 \equiv 2 \pmod 8$ and $(A-1)^2+1$ is odd, and if A is even the roles are exchanged. So no condition modulo 2 is required and $2^7 \nmid f(A)$ always. *(ii)* If $q \equiv 3 \pmod 4$ then $A^2+1 \equiv 0 \pmod q$ has no solution, so $q \nmid f(A)$; this removes $3, 7, 11, 19, 23, \ldots$ at once. *(iii)* In particular $7 \nmid f(A)$ for every A, so *no selected bad place has residue characteristic* 7. [P] *This is not the same as the order condition of [IUT1, Def. 3.1(c)]*, which asks that ℓ not divide $v_{\mathfrak{p}}(a)$; that is secured at the places away from π by seventh-power-freeness, and at π itself only by the extra requirement $7 \nmid N$, since the Tate order there is $2N$. *These are orders at places of L, and [IUT1, Def. 3.1(c)] asks about places of F*: for $w \mid v$ the Tate order is multiplied by $e(w/v)$, and since $F = L(E_0[15])$ is Galois over L with $[F:L] \mid |\mathrm{GL}_2(\mathbb{Z}/15)| = 23040 = 2^9 3^2 5$, every $e(w/v)$ is prime to 7. So an order prime to 7 downstairs stays prime to 7 at every bad place of F. The second congruence makes $v_5(f(A)) = 0$.

Hence a prime q with $q^7 \mid G_N(x)$ satisfies $q \equiv 1 \pmod 4$ and $q \geq 13$. *This is a statement about seventh powers, not about divisors*: 2 divides $G_N(x)$, since $v_2(f(A)) = 1$ always. Such a q is neither 5 nor p, by the two paragraphs above, and is not 2; *therefore* $q \nmid M = 5p^{N+1}$, so multiplication by M is invertible modulo q^7 and the roots in x correspond bijectively to the roots of f. Such q are prime to $\mathrm{disc}(f) = 400$, so f has at most four simple roots modulo q, each lifting uniquely modulo q^7. With $0 < G_N(x) < C_N(X+1)^4$ for $1 \leq x \leq X$ and $C_N = M^4/p^N$, the number of excluded x is at most

$$\sum_{\substack{q \text{ prime},\ 13 \leq q \leq Q_N \\ q \equiv 1\ (4)}} 4\Big(\frac{X}{q^7} + 1\Big) \leq 7.38 \times 10^{-8}\, X + 4C_N^{1/7}(X+1)^{4/7}, \qquad Q_N := \big\lfloor (C_N(X+1)^4)^{1/7} \big\rfloor,$$

whose second term is $o(X)$ for fixed N and whose first coefficient is far below 1. Infinitely many x therefore remain. *The constant is proved, not estimated*: summing $4q^{-7}$ exactly over the primes $q \equiv 1 \pmod 4$ with $13 \leq q \leq 100$ and bounding the rest by $4\int_{100}^{\infty} x^{-7}\,dx = 2/(3 \cdot 10^{12})$ — which over-counts, since it runs over all integers — gives $7.3794978287\ldots \times 10^{-8} < 7.38 \times 10^{-8}$ in exact rational arithmetic (`scripts/imq_family.py`). *A truncated positive sum is only a lower bound*, so the earlier floating-point check did not establish this inequality.

[P] *The range matters*: written over all q the term $+1$ would sum to infinity. Only $q \leq Q_N$ can produce an exclusion, so the $+1$ contributions are counted over that finite range; the main terms X/q^7 are then extended to the full sum, which only increases them. The density bound is unaffected. $\square$

This is an arithmetic existence statement. It does not verify (d)–(f) for each member; those are the geometric conditions of §4, established for the datum (1).

6 Three ways in which $\mathbb{Q}(i)$ is simpler

[P] Compared with the datum of [First] over $\mathbb{Q}(\sqrt{5})$:

(i) *The node is already split.* Over $\mathbb{Q}_{211}$ the element -1 is a nonsquare, so [First] adjoins i to split the bad node. Here $109 \equiv 1 \pmod 4$ and $i \in F_{\mathrm{mod}}$, so both nodal types split over L itself (§4).

(ii) *The degree bound is halved.* Write $G = |\mathrm{GL}_2(\mathbb{Z}/15\mathbb{Z})| = 23040$. Here $[F : F_{\mathrm{mod}}]$ divides G, whereas in [First] it divides $2G = 46080$: no factor of 2 is spent on adjoining i.

(iii) *The family needs two congruences instead of five.* The five conditions of the family in [First] reduce here to the two of Proposition 2, and $\mathrm{disc}(f) = 400$ rather than $144400 = 20^2 19^2$. Comparing the two excluded densities *as the same quantity* — the sum $\sum 4/q^7$ over the primes each family must exclude, not the integral majorant $1/69984$ that [First] uses in its counting step — the value falls from 4.93×10^{-6} to 7.38×10^{-8}, a factor of 66.8. One term accounts for almost all of it: every $q \equiv 3 \pmod 4$ is excluded here automatically, *including* $q = \ell = 7$, and $4/7^7 = 4.86 \times 10^{-6}$ is 98.5% of the density for [First]. So the structural gain and the numerical gain are the same fact: the condition on ℓ becomes automatic rather than a separate check.

7 Roomier variants

[P] The constraint that keeps I^* empty is $e_{\mathrm{b}} \le p - 2$ with

$$e_{\mathrm{b}} = \frac{15\ell}{\gcd(15\ell, 2N)},$$

p split in F_{mod}. *It reduces to* $e_{\mathrm{b}} = 15\ell$ *only when the gcd is* 1, as it is at $(N, \ell) = (29, 7)$, where $\gcd(105, 58) = 1$; the figures below are for that fixed pair. Varying N changes the gcd — at $(114, 37)$ it is 3 — and also requires $\ell \nmid 2N$ to be rechecked. For $\ell = 7$ and $N = 29$ this asks $p \ge 107$; the datum above uses the first such prime, so the slack $p - 2 - e_{\mathrm{b}}$ is 2. Larger split primes give more: 113, 137, 149, 157, 173, 181, 193 leave $6, 30, 42, 50, 66, 74, 86$. Every imaginary quadratic field of class number one admits a split prime ≥ 107. We have kept $p = 109$, and $N = 29$, so that the local invariants agree exactly with [First]. *N is not free*: moving it changes $\gcd(15\ell, 2N)$, hence e_{b}, and the condition $\ell \nmid 2N$ of [IUT1, Def. 3.1(c)] has to hold as well.

8 The stage margins and their thresholds

[P] With $I^* = \varnothing$ admissible in [IUT4, Prop. 1.4(iii)], and *for a tuple whose last entry is bad* — the case in which $\lambda = j^2 s$ rather than 0, by Step (v) of the proof of [IUT4, Thm. 1.10] — the difference co − bd at stage j is $j^2 s - (j+2)$, and it is positive exactly when $\mathrm{ord}_p(q) > 2\ell(j+2)/j^2$. *These are evaluations of the coefficients in the bound*, not a comparison of the actual Θ-pilot with anything.

j	$j^2 s$	$j^2 s - (j+2)$	positive iff $\mathrm{ord}_p(q) >$
1	29/7	8/7	42
2	116/7	88/7	14
3	261/7	226/7	70/9

[P] *The binding stage is $j = 1$, by a factor of three over $j = 2$ and of more than five over $j = 3$.* Everything about the threshold is therefore decided at the first stage.

9 Where the data on record sit

[P] $\mathrm{ord}_p(q) = -v_p(j_E)$ at a place of multiplicative reduction. At $\ell = 7$ the stage-one threshold of §8 reads $\mathrm{ord}_p(q) > 42$.

datum	p	$\mathrm{ord}_p(q)$	s	$s-3$
[DH25], Cremona 11a1 over $\mathbb{Q}$	11	5	5/14	−2.64
Cremona 14a1 over $\mathbb{Q}$	7	3	3/14	−2.79
Cremona 15a1 over $\mathbb{Q}$	5	4	2/7	−2.71
Cremona 37a1 over $\mathbb{Q}$	37	1	1/14	−2.93
[First] over $\mathbb{Q}(\sqrt{5})$	211	**58**	29/7	**+1.14**
the present datum over $\mathbb{Q}(i)$	109	**58**	29/7	**+1.14**

[P] *The last column of the four $\mathbb{Q}$-rows is the value of the formula, not a licensed comparison*: at 14a1 $p = \ell$ (which [IUT1, Def. 3.1(c)] does not allow at a bad place at all), and at 15a1 $p \mid 15$. At 11a1 and 37a1 the exclusion is by size, and *the bound does not depend on choosing a field*: [IUT1, Def. 3.1(b)] requires the $2 \cdot 3$-torsion of E_F to be rational over F, so at the reference level $\ell = 7$ the field K contains $E[42]$. Both primes are prime to 42, and their Tate orders are 5 and 1, so

$$e_{\mathrm{b}} \;\geq\; \frac{42}{\gcd(42, \mathrm{ord}_p(q))} = 42 \quad \text{at both}, \qquad 42 > 9, \qquad 42 > 35.$$

An earlier version printed $105/\gcd(105,5) = 21$ *at 11a1 and* 105 *at 37a1*, carrying over the torsion level 105 of the datum (1) to curves that do not have it; [DH25] takes $F = \mathbb{Q}(i, E[30])$, which at level 7 would give $210/\gcd(210,5) = 42$ and at its own level 13 gives $390/\gcd(390,5) = 78$. The exclusions are unchanged; the printed indices were not the indices of these curves. They are printed to show the scale of $\mathrm{ord}_p(q)$ on record, nothing more.

[P] The sample above is not the whole record. Sweeping the Cremona tables of conductor below 10^4 — 64687 curves — for a *single* place of multiplicative reduction with $\mathrm{ord}_p(q) > 42$ returns 80 curves. Of these, 78 have that place above 2, which [IUT1, Def. 3.1] excludes: bad places of residue characteristic two are not admitted. The remaining two, Cremona `5160j1` and `5190c3`, have the place above 3 with $\mathrm{ord}_3(q) = 44$, so that at $\ell = 7$

$$s = \tfrac{44}{14} = \tfrac{22}{7}, \qquad s - 3 = \tfrac{1}{7} > 0.$$

[P] So the threshold *is* reached inside the tabulated range, and it is reached by curves that have been on record since long before this line of work. What those two curves do not do is admit the simplified form of the estimate: with $p - 2 = 1$ and $e \geq 2$ (see below), those places must be put into the set I^* of [IUT4, Prop. 1.4(iii)]. *The estimate still applies*: Step (v) of [IUT4, Thm. 1.10] is written for exactly this case and absorbs such places into $4(j+1)\,\ell^*_{\mathrm{mod}}$. What is not available at those curves is the $I^* = \varnothing$ form, which is the one the margin of §8 is computed from; the margin is therefore not the applicable quantity there.

[P] *No number is put on e at those two curves.* The prime-to-p expression $15\ell/\gcd(15\ell, \mathrm{ord}_p(q))$ is not available when p divides the torsion level, and here $p = 3$ divides 105. What can be said without it is enough: full 3-torsion forces $\mu_3 \subset K$ by the Weil pairing, and $\zeta_3 - 1$ is a root of the Eisenstein polynomial $T^2 + 3T + 3$, so $\mathbb{Q}_3(\mu_3)/\mathbb{Q}_3$ is totally ramified of index 2 and $e \geq 2 > 1 = p - 2$. *This also settles the number that the unavailable expression would have returned*: $2 \mid e$, so $e = 105$ is not merely unproved at those places but impossible. The threshold of §8 is therefore not by itself a useful filter; it is one of five conditions, and §12 treats them together. The two data of [First] and above were built with $a = \pi^{29}$, which forces $\mathrm{ord}_p(q) = 58$ at a prime large enough for $e_{\mathrm{b}} \le p - 2$ to hold as well.

10 Tuple types

[S] [IUT4, Thm. 1.10, Step (v)] fixes λ by the status of the *last* index alone, verbatim: "λ" to be 0 if $\underline{v}_j \in \underline{\mathbb{V}}^{\mathrm{good}}$; "$\lambda$" to be "ord(−)" of the element $\underline{\underline{q}}_{\underline{v}_j}^{j^2}$ [...] if $\underline{v}_j \in \underline{\mathbb{V}}^{\mathrm{bad}}$. *The q carries the double underline*: it is the 2ℓ-th root of [IUT1, Ex. 3.2(iv)], not q itself, which is why $\lambda = j^2 s$ with $s = \mathrm{ord}_p(\underline{\underline{q}}) = 29/7$ and not $j^2\,\mathrm{ord}_p(q) = 58j^2$. [P] Write $d_I = (j+1) - \sum_i e_{t_i}^{-1}$, and put

$$\mathrm{co}(t) := -\sum_i e_{t_i}^{-1}, \qquad \mathrm{bd}(t) := -\lambda + d_I + 1.$$

$\mathrm{co}(t)$ *is the coefficient expression, not the log-volume of the tensor packet*, and the two differ once a tuple has more than one bad entry. The good side is unramified, so a factor R_{g} costs nothing. With k bad entries R_I is non-maximal *exactly when* $k \ge 2$; for $k = 0, 1$ it is maximal. The packet $\log_p(R_I^\times)$ therefore has normalized log-volume $\mathrm{co}(t)\log p + \mu^{\log}(R_I)$, that is, *below* $\mathrm{co}(t)\log p$ by

$$-\mu^{\log}(R_I) = \tfrac{\max\{k-1,0\}(e-1)}{2e}\log p, \qquad e = e_{\mathrm{b}} = 105,$$

which is $\frac{52}{105}\log p$ for $k = 2$ — the gain recorded in §15 — and $\frac{104}{105}\log p$ for $k = 3$. So the rows unaffected are those with *at most one* bad entry; among the rows printed below, (b, b), (g, b, b) and (b, b, g) carry the $k = 2$ defect and (b, b, b) the $k = 3$ one. *Nothing in the table depends on this*: the columns are co and bd as defined. (Earlier drafts called co the "seed", which collided with the $\lceil\lambda\rceil$ of §16; the two are different quantities and are now named apart.)

j	tuple	$\mathrm{co}(t)$	$\mathrm{bd}(t)$	co − bd
1	(g, g)	−2	1	−3
1	(g, b)	−106/105	−226/105	**8/7**
1	(b, g)	−106/105	209/105	−3
1	(b, b)	−2/105	−122/105	**8/7**
2	(g, g, g)	−3	1	−4
2	(g, b, b)	−107/105	−1427/105	**88/7**
2	(b, b, g)	−107/105	313/105	−4
2	(b, b, b)	−1/35	−63/5	**88/7**

[P] *The difference is constant on each of two classes, not on all types.* It is 8/7 at $j = 1$ and 88/7 at $j = 2$ exactly for the tuples whose last entry is bad, and −3, −4 for those whose last entry is good. In particular "mixed" does not name a value: (g, b) and (b, g) have the same co and different bounds. The cancellation behind the constant value on the first class is $d_I + \sum_i e_{t_i}^{-1} = j + 1$; see [First, §6].

Remark 4 (correction)**.** [P] An earlier version of this table set $\lambda = j^2 s$ for *every* type, including the all-good ones, and reported a single "mixed" row. Both are corrected above. The all-good rows change from (−22/7, 8/7) and (−109/7, 88/7) to (1, −3) and (1, −4); the rows for tuples ending in a bad place are unchanged.

11 How much probability each tuple carries

[S] [DHPS] puts a probability measure on the tuples, and the tables above are per-tuple quantities, so it is worth recording what weight each of them receives on the present data. The measure is stated twice; we quote both as printed. In [DHPS, §1, p. 2] the numerator is the local degree of the *base* field, $\Pr(v) = [F_{0,v} : \mathbb{Q}_p]/[F_0 : \mathbb{Q}_p]$; in [DHPS, §7.1, p. 22] the denominator is $[F_0 : \mathbb{Q}]^{j+1}$. Read together the two statements fix the measure, and the degree formula $\sum_{v|p}[F_{0,v} : \mathbb{Q}_p] = [F_0 : \mathbb{Q}]$ then makes it a probability measure:

$$\Pr(v_0, \ldots, v_j) = \frac{2}{\ell - 1} \prod_{i=0}^{j} \frac{[F_{0,v_i} : \mathbb{Q}_p]}{[F_0 : \mathbb{Q}]}.$$

[P] For both data of §2 the prime splits in F_0, so there are two places above it, each of local degree 1, and $[F_0 : \mathbb{Q}] = 2$. With $\ell = 7$ the index j runs over $1, 2, 3$:

j	tuples 2^{j+1}	Pr per tuple	mass at this j
1	4	1/12	1/3
2	8	1/24	1/3
3	16	1/48	1/3
	28		1

[P] So the mixed $j = 1$ tuple of §10 — the one carrying the 8/7 of that table — carries 1/12 of the mass, and each index j contributes 1/3 regardless of how many tuples it has. Two remarks on reading the source here. *We adopt a reading for the sole purpose of evaluating the displayed formula, and make no claim about what either point does to the arguments of that paper.* The numerator in [DHPS, §7.1] is printed with a completion degree of K rather than of F_0; taken literally the tuples would carry masses well above 1, and it is the §1 statement that identifies the field. We do not consider what either point does to the estimates of that paper; we record the reading only because the tables here evaluate that very formula on explicit data, so the normalisation has to be pinned down, and we use it as the degree normalisation of [IUT4, Rem. 1.7.1].

12 The region cut out by five conditions

Five conditions can be asked of a datum in this family. *Only (2) and (5) are hypotheses the source imposes.* (1) is the threshold derived in §8; (3) is a non-vacuity criterion of ours; and (4), as explained below, is not a hypothesis at all but the condition under which a place may be left out of the set I^* of [IUT4, Prop. 1.4(iii)].

(1) $\mathrm{ord}_p(q) = 2N > 6\ell$ — *the stage-one threshold, §8*

(2) $\sqrt{H_\forall} \le \ell \le 10\delta\sqrt{H_\forall}\log(2\delta H_\forall)$ — [S] [IUT4, Cor. 2.2(ii), (C1)], *both sides*

(3) $\dfrac{\ell+1}{24} N \log p > (\ell + 5)\log(e^*_{\mathrm{mod}}\ell)$, both sides in nats — [P] from [IUT4, Thm. 1.10, Step (v)]

(4) $e_{\mathrm{b}} = 15\ell/\gcd(15\ell, \mathrm{ord}_p(q)) \le p - 2$ — [P] $I^* = \varnothing$ in [IUT4, Prop. 1.4(iii)]

(5) $\ell \nmid \mathrm{ord}_p(q)$ — [S] [IUT1, Def. 3.1(c)]

[P] *Condition (4) is not a restriction on the source's estimate, and it is worth being exact about this.* [IUT4, Prop. 1.4(iii)] reads "Let $I^* \subseteq I$ be a subset such that for each $i \in I \setminus I^*$, it holds that $p - 2 \geq e_i$", so I^* is *chosen* and the inequality is required only outside it; the bound then carries $4|I^*|/p$ and $\sum_{i \in I^*}\{3 + \log(e_i)\}$. [IUT4, Thm. 1.10, Step (v)] takes I^* to be exactly the set of i with $e_{v_i} > p_{v_\mathbb{Q}} - 2$, and by (R4) absorbs them into $4(j+1)\,\ell^*_{\mathrm{mod}}$ — the same term tabulated in §17. So the estimate applies whether or not (4) holds; what (4) buys is the error-free form, with $I^* = \varnothing$.

[P] *Two heights occur here and must be kept apart.* Condition (C1) of [IUT4, Cor. 2.2(ii)] is stated for $\log(q^\forall)$, which keeps every nonarchimedean place; write $H_\forall$ for it. The height over the *selected* bad places, which drops those over 2 and ℓ, is a different number; write H_{sel}. For $a = \pi^N$ over $\mathbb{Q}(\sqrt{5})$, with $B = \mathrm{N}(1-a)$,

$$H_\forall = N \log p + \log B - \min(v_2(B), 8) \log 2, \qquad H_{\mathrm{sel}}(\ell) = N \log p + \log B - v_2(B) \log 2 - v_\ell(B) \log \ell.$$

[P] H_{sel} *drops the place over that row's own* ℓ*, not over a fixed prime.* The two heights agree only at $N = 29$, where $\ell = 7$ and $v_7(B) = 0$; at $N = 114$, $\ell = 37$ one has $v_{37}(B) = 2$ and they differ by $2 \log 37$. *Condition (2) is tested against* $H_\forall$; both are printed below.

N	ℓ	$H_\forall$	$\sqrt{H_\forall}$	e_{b}	(1)(2)(3)(4)(5)	H_{sel}
29	7	309.02	17.58	105	`o X X o o`	309.02
114	**37**	1217.45	34.89	185	`o o o o o`	1210.23
160	53	1707.05	41.32	159	`o o o o o`	1704.28
480	73	5132.24	71.64	73	`o o o o o`	5129.47
480	79	5132.24	71.64	79	`o o o o o`	5129.47

[P] *Of the rows listed,* $N = 114$, $\ell = 37$ *is the first that passes all five*, at the same prime $p = 211$ and in the same family as [First]. *The conditions are not monotone in* N *and do not stay satisfied*: in the same family $N = 116$, $\ell = 37$ gives $e_{\mathrm{b}} = 15 \cdot 37/\gcd(232, 555) = 555 > 209$ and fails (4).

[P] *Condition (5) has two parts and they are not on the same footing.* At the designated place the order is $\mathrm{ord}_p(q) = 2N$, and $\ell \nmid 2N$ is an integer check. At the remaining bad places it asks that $v_\mathfrak{p}(1-a)$ be prime to ℓ, for which freedom of $\mathrm{N}(1-a)$ from ℓ-th powers is *sufficient but not necessary* — a valuation of $\ell+1$ is prime to ℓ and not ℓ-th-power-free. For the datum (1) of the present report that computation is shipped: `scripts/imq_sieve7.cpp` runs the exhaustive sieve to $\lfloor C^{1/7} \rfloor$ (it is not the sieve for the datum of [First]). *For the larger data of §12 and §13 it is not*: the cyclotomic factorization $a - 1 = \prod_{d|N} \Phi_d(\pi)$ reduces the search, but on its own it excludes an ℓ-th power only for factors whose norm is small enough, and the certificate for the reduced search is carried out by `scripts/powerfree_certificate.py`, which ships with this report and runs in well under a second. The rows above should be read accordingly.

Remark 5 (corrections)**.** **[P]** An earlier version of this table compared the two sides of condition (3) *in different units* — the left in $\log p$, the right in nats — which put `X` in column (3) for $N = 114$ and $N = 160$ and made $N = 480$ look like the first point at which all five hold. With both sides in nats — equivalently, dividing both by $\log p > 0$, which is the form the code uses — the first of the rows listed at which all five hold is $N = 114$ — not every N beyond it: $N = 116$, $\ell = 37$ fails (4). The same earlier version used $e_{\mathrm{b}} = 15\ell$ without the gcd, omitted condition (5), and reported a sieve analysis in a form the cyclotomic factorisation replaces by a finite blockwise one. All four are corrected here. The rows $N = 114$ and $N = 160$ are due to an external computation, which also supplied the certified values of H used above for calibration.

13 A curve carrying fourteen candidate levels

[P] The five conditions of §12 do not act independently. Raising N helps (1) and (3) but raises H, which by (2) forces ℓ up, which by (4) forces p up. For a datum of the shape $a = \pi^N$ the tension resolves into a *window*: once (3), (4) and (5) hold, and once the upper half of (C1) is checked, the admissible ℓ are the primes with

$$\sqrt{H_\forall} \ \le\ \ell\ <\ \tfrac{N}{3},$$

the left end being the *lower* half of the source's condition [IUT4, Cor. 2.2(ii), (C1)] and the right end the threshold derived in §8 — *the latter is ours, not the source's. Condition (C1) is two-sided, and the window is not by itself equivalent to it*: we check the upper half separately at each datum tabulated, and we do *not* show that $\ell < N/3$ implies it in general. At the datum below that upper half, $\ell \le 10\delta\sqrt{H}\log(2\delta H)$ with $\delta = 2^{12}\cdot 3^3\cdot 5\cdot d$, where d *bounds* the degree in $x_E \in U_X(\overline{\mathbb{Q}})^{\le d}$ [IUT4, Cor. 2.2], is not binding: the smallest admissible value here is $d = 2 = [\mathbb{Q}(a):\mathbb{Q}]$, giving $\delta = 1105920$ and a bound of about 2.2×10^{10}, and the bound only grows with d; against a window whose right end is $157.33\ldots$. [IUT1, Def. 3.1(c)] additionally requires $\ell \ge 5$ and ℓ prime to the residue characteristics of the bad places. The bad places lie above p *and* above every prime dividing $\mathrm{N}(1-a)$, so the second condition is not settled by $\ell < p$; what settles it is that $\ell \nmid \mathrm{N}(1-a)$, which we checked for each of the fourteen. A datum is therefore interesting to the extent that this window is wide.

[P] Take

$$L = \mathbb{Q}(i), \quad \pi = 41 + 26i, \quad \mathrm{N}(\pi) = 2357, \quad a = \pi^{472}, \quad E : y^2 = x(x-1)(x-a),$$

with $F = L(E[15])$ and $K_\ell = F(E[\ell])$. Here 2357 is prime and $\equiv 1 \pmod 4$, so it splits in L, and one place above it is multiplicative and one good. Then $\mathrm{ord}_p(q) = 944$, the finite Tate height is $H = 7324.7516\ldots$, and the window $85.58 \le \ell < 157.33$ contains *fourteen* primes:

ℓ	e_{b}	$e_{\mathrm{b}} \le p-2$	margin $s-3$ at $j=1$
89	1335	yes	205/89
97	1455	yes	181/97
101	1515	yes	169/101
103	1545	yes	163/103
107	1605	yes	151/107
109	1635	yes	145/109
113	1695	yes	133/113
127	1905	yes	91/127
131	1965	yes	79/131
137	2055	yes	61/137
139	2085	yes	55/139
149	2235	yes	25/149
151	2265	yes	19/151
157	**2355**	**yes**	**1/157**

[P] At $\ell = 157$ *two upper constraints are simultaneously tight*: 1/157 is the last positive stage-one margin, since $472/3 = 157.33\ldots$; and $e_{\mathrm{b}} = 15\cdot 157 = 2355 = p-2$ exactly. *Both are upper constraints*; the left wall of the window, $\sqrt{H_\forall} = 85.58\ldots$, is what makes 89 the first admissible prime and has nothing to do with 157.

[P] *What the fourteen are, exactly.* They are the ℓ for which the five numerical filters of §12 hold at this curve. *They are not fourteen verified initial* Θ*-data*: no theorem here applies [IUT1, Def. 3.1](a)–(f) to each of them, the basis-free image argument of §3 is carried out for the datum of (1) and not for these, and the order condition at the non-designated bad places is in the state described in §12. Two further remarks. *It is fourteen levels on one curve, not fourteen curves*: F_{mod}, E and p are fixed and only ℓ moves. And the order condition of [IUT1, Def. 3.1(c)] at every bad place is what costs the work here. $\mathrm{N}(1-a)$ has 1592 digits, so a direct 157-th power bound runs to about $10^{10.14}$. The reduction uses *nine blocks*: the cyclotomic factors $\Phi_1, \Phi_2, \Phi_4, \Phi_8, \Phi_{59}, \Phi_{118}, \Phi_{236}$ kept whole, together with

$$G_+ = \frac{X^{118}+i}{X^2-i}, \qquad G_- = \frac{X^{118}-i}{X^2+i}, \qquad G_+G_- = \Phi_{472},$$

whose degrees $1, 1, 2, 4, 58, 58, 116, 116, 116$ sum to 472. *It is* Φ_{472} *that splits, not* Φ_4. Clearing the $\binom{9}{2} = 36$ pairwise gcds then reduces the problem to a sieve small enough to run exhaustively, which is why the condition can be settled for all fourteen levels at once rather than one at a time. *The digit count, the factor count and the gcd count are checked by* `scripts/tables.py`; *The sieve is shipped and runs*: `scripts/powerfree_certificate.py` rebuilds the nine blocks from π by exact Gaussian division — each $\Phi_d(\pi)$ from $\prod_{e|d}(\pi^e - 1)^{\mu(d/e)}$, with the product of all nine asserted equal to $\pi^{472}-1$ — then sieves each block norm exhaustively to its own 89th-root bound, the largest being 24,856, and finds no 89th power; 8,318 primes are tested in all. It then checks the only way blockwise freeness could fail to give freeness of B: the 36 pairwise gcds show that just 2 and 5 meet two blocks, and $v_2(B) = 9$, $v_5(B) = 2$, both below 89. Output in `scripts/powerfree_certificate_output.json`; the same constants are recorded in `lean/LargeImaginaryDatum.lean`, whose header states that having them does not by itself prove power-freeness. *This establishes the sufficient condition, not fourteen initial* Θ*-data.*

14 Coefficients printed in the source

[S] Step (v) of the proof of [IUT4, Thm. 1.10] carries the coefficient $j^2/(2\ell)$ on $\log(q)$ — the *unrooted* q, since $\underline{\underline{q}} = q^{1/(2\ell)}$ makes $\log(q^{j^2/(2\ell)}) = j^2 \log \underline{\underline{q}}$, so on $\log \underline{\underline{q}}$ the coefficient would be j^2 — and prints the average over $j \in \{1, \ldots, \ell^*\}$ first as $(2\ell^*+1)(\ell^*+1)/(12\ell)$ and then as $(\ell+1)/24$. At $\ell = 7$:

$$\tfrac{1}{14},\ \tfrac{2}{7},\ \tfrac{9}{14} \quad \text{average } \tfrac{1}{3}, \qquad q\text{-side } \tfrac{1}{2\ell} = \tfrac{1}{14}, \qquad \text{ratio } \tfrac{12}{\ell(\ell+1)} = \tfrac{3}{14}.$$

[P] The two printed forms agree, and the average is $\ell(\ell+1)/12$ times the stage-one coefficient.

15 Local degrees and tensor orders

[P] This section is computed for the datum of [First] at 211; the corresponding numbers for the datum of (1) at 109 differ, as noted in the abstract. At 211 the good completion is unramified of degree 24: the reduction has 188 points, trace 24, and $X^2 - 24X + 211$ has Frobenius of order $4, 3, 8$ modulo $3, 5, 7$, with least common multiple 24. The bad completion has $e = 105$, $f = 2$. So $a_{\mathrm{g}} = 24$, $a_{\mathrm{b}} = 210$, and the tensor orders are

$$N_{(\mathrm{g},\mathrm{g})} = 576, \qquad N_{(\mathrm{g},\mathrm{b})} = 5040, \qquad N_{(\mathrm{b},\mathrm{b})} = 44100.$$

[P] *The mixed tensor order is maximal and the all-bad one is not.* Because the good side is unramified, $\mathcal{O}_{\mathrm{g}} \otimes \mathcal{O}_{\mathrm{b}}$ is the full ring of integers, with normalized gain 0; for (b, b) the gain is

$(e-1)/(2e) = 52/105$ in units of $\log p$. [S] [IUT3, Prop. 3.2] — which is what [IUT4, Thm. 1.10] cites here, using [IUT3, Thm. 3.11(i)(a)] only for the notation — records that *each of the* $j+1$ *factors, before tensoring*, is a direct sum indexed by the places of $\underline{\mathbb{V}}$ over $v_{\mathbb{Q}}$ — 2 of them here, 1 over $F_{\mathrm{mod}} = \mathbb{Q}$. *This is not the number of tuple components after tensoring* (that is 2^{j+1}), nor the number of field factors into which a local tensor product splits.

16 Containers, seeds, and the floor slack

[S] Step (v) works inside a container whose exponent is $\lfloor \lambda \rfloor - |I|$. *The exponent alone does not name a set*: in [IUT4, Prop. 1.4(iii)] it multiplies $\log_p(R_I^{\times})$, and the table below records exponents, not lattices. "Seed" in that table means the same thing rounded the other way: $\lceil \lambda \rceil$ is the least integer exponent of the same lattice $\log_p(R_I^{\times})$ that is at least λ, so the two columns are the upward and downward discretisations of one real number against one reference lattice. And the bound of [IUT4, Prop. 1.4(iii)] takes a floor, leaving a slack.

[P] *The table is restricted to tuples whose final entry lies in* $\underline{\mathbb{V}}^{\mathrm{bad}}$, so that $\lambda = j^2 s$. If the final entry is good then $\lambda = 0$ by Step (v) of [IUT4, Thm. 1.10], and none of the numerical columns below applies.

j	$\lambda = j^2 s$ ($\underline{v}_j \in \underline{\mathbb{V}}^{\mathrm{bad}}$)	exponent $\lfloor \lambda \rfloor - \|I\|$	seed $\lceil \lambda \rceil$	floor slack
1	29/7	2	5	6/7
2	116/7	13	17	3/7
3	261/7	33	38	5/7

17 The slack in the averaged bound

[S] Step (v) of the proof of [IUT4, Thm. 1.10] carries, besides the terms of [IUT4, Prop. 1.4(iii)], a term $4\iota_{v_{\mathbb{Q}}} \ell^*_{\mathrm{mod}}$ with $\ell^*_{\mathrm{mod}} = \log(e^*_{\mathrm{mod}} \ell)$ and $\iota = 1$ when $p \le e^*_{\mathrm{mod}} \ell$; here $e^*_{\mathrm{mod}} = 2^{12} 3^3 5\, e_{\mathrm{mod}}$ with $e_{\mathrm{mod}} \le d_{\mathrm{mod}} = 2$.

[P] For the datum of [First], $e^*_{\mathrm{mod}} \ell = 7\,741\,440$, so $\iota = 1$ at $p = 211$ and $\ell^*_{\mathrm{mod}} = 2.964 \log p$. After the procession-normalized average the term contributes $(\ell+5)\ell^*_{\mathrm{mod}} = 35.57 \log p$, whereas the entire q-signal at 211 is $\frac{\ell+1}{24} \cdot 29 = 9.667 \log p$:

$$\boxed{\frac{(\ell+5)\,\ell^*_{\mathrm{mod}}}{\frac{\ell+1}{24} \cdot \frac{\mathrm{ord}_p(q)}{d_{\mathrm{mod}}} \cdot \log p} = 3.6792605719\ldots}$$

[P] *The* q*-term is written with the unrooted* q: here $\mathrm{ord}_p(q) = 58$ and $d_{\mathrm{mod}} = [F_{\mathrm{mod}} : \mathbb{Q}] = 2$, so $\frac{\ell+1}{24} \cdot \mathrm{ord}_p(q)/d_{\mathrm{mod}} = 29/3$. *That this equals* $\frac{\ell+1}{24}\, \mathrm{ord}_p(\underline{\underline{q}})\, \ell$ *is particular to* $d_{\mathrm{mod}} = 2$, since $\mathrm{ord}_p(\underline{\underline{q}}) = \mathrm{ord}_p(q)/(2\ell)$. *Inside the averaged bound, the* ι*-term exceeds the* q*-term by a factor of nearly four.* This compares two terms of one upper bound; it says nothing about the distance from that bound to the quantity it bounds, and it is not a criticism of the bound. Note also that at $p = 211$ one has $e_{\mathrm{b}} = 105 \le 209$, so $I^* = \varnothing$ is available here and the ι-term is the price of the *uniform* form, not something forced by this datum.

Remark 6 (correction)**.** [P] An earlier version printed this ratio without the $\log p$ in the denominator, where the numerator is in nats and the denominator in units of $\log p$; as printed it evaluates to $19.6908\ldots$, not 3.68. The intended quantity, with both in nats, is the boxed one.

18 What the numbers suggest, with the reservations attached

[P] The tables above are meant to stand on their own, and each is checked by `scripts/tables.py`. We nonetheless set out what they appear to indicate, as four readings, each paired with the reservation that limits it. *Nothing above depends on any of them*; where a reading is echoed elsewhere — (R1) in the abstract, (R2) in §9, (R3) in §17 — it is as a remark, never as a step.

(R1) **The admissible region is thin.** For data of the shape $a = \pi^N$, and for a candidate at which conditions (3), (4) and (5) of §12 already hold, the two remaining conditions reduce to $\sqrt{H_\forall} \leq \ell < N/3$ *together with the upper half of* (C1) — which is not binding at any datum tabulated here, and which we have not shown to be non-binding in general (§13). So the room available to numerical work on these estimates is set by [IUT4, Cor. 2.2(ii)] from below and by the stage-one threshold from above. *Reservation:* the window is not by itself the admissible region — $N = 116$, $\ell = 37$ lies inside it and fails (4), with $e_{\mathrm{b}} = 555 > 209$ — and this is computed for one shape of datum in two families. A datum not of the form $a = \pi^N$ need not give a window of this kind, and we have not looked.

(R2) **The entry threshold is not, by itself, a criterion.** Two curves in the tabulated range clear it while requiring $I^* \neq \varnothing$ (§9), so "does the stage-one comparison say anything?" is not settled by the threshold alone. *Reservation:* the sweep covers Cremona conductor below 10^4 and a single multiplicative place. Larger conductors, and configurations with several bad places, are untouched.

(R3) **At these parameters the ι-term of the averaged bound dominates its q-term**, by a factor of about 3.7 (§17). *Reservation:* this compares two terms *inside* an upper bound. It is not a statement about the distance between that bound and the true value — which is what tightness would mean — and it says nothing whatever about whether the estimate holds. A bound may be far from tight and perfectly correct — indeed [IUT4, Thm. 1.10] is stated for an arbitrary datum, so slack at a particular one is what one should expect.

(R4) **These estimates can be evaluated on explicit input.** Fourteen candidate levels on one curve, together with the data of [DH25] over $\mathbb{Q}$ and the family of Proposition 2, give enough explicit input for the local quantities to be computed rather than discussed. *Reservation:* constructing a datum and evaluating what the theory does with it are different tasks. Ordinary normalised log-volumes of the lattices involved, coefficients, and ratios of terms in the explicit bounds are computed here; what is *not* computed is the hull of the union of the possible images of the actual Θ-pilot, either side of the comparison of [IUT3, Cor. 3.12], or membership in the exceptional sets that the effective statements exclude.

[P] One reading we deliberately do *not* offer: that any of this bears on whether [IUT3, Cor. 3.12] holds. The quantities tabulated here are inputs to the estimates, not their outputs, and no step of the present report examines the inference those estimates rest on.

Remark 7 (what this report draws on)**.** [P] A report consisting of tables can read as an audit of the papers it draws on. It is not one, and the reason is worth stating precisely. The conditions in §12 are of two kinds, and we keep them apart: (2) is [IUT4, Cor. 2.2(ii)] and (4) is the condition under which [IUT4, Prop. 1.4(iii)] may be applied with $I^* = \varnothing$; (5) is [IUT1, Def. 3.1(c)]; and (1) and (3) are conditions imposed here. The source's own conditions can be tabulated at all only because they are stated precisely enough to be checked against a named curve, and several errors in earlier drafts of this report were caught by those same statements. The explicit data of [DH25] are what these questions are evaluated against. The programme described in [LANA] is a separate formalization effort that may prove complementary; we do not claim that its targets coincide with the interface assumed here. The exchange recorded in [SS] and [Rpt] located the point at which the inference has to be understood; this report does not enter it and takes no position on it. The distinction that governs everything above is between computing with a definition and understanding a proof: only the former is attempted here.

19 Scope

Claimed. (1) is an initial Θ-datum with a mixed local profile above 109 and with F_{mod} imaginary quadratic; the family of Proposition 2 exists.

Not claimed. That [MFHMP, Thm. A] is affected or that any estimate is; that the *abc* conjecture or any statement of [IUT1, IUT2, IUT3, IUT4] is false; that the hypothesis (SA) of [First] holds; that the analysis of [SS], the responses of [Rpt] or the findings of [LANA] bear on the datum. Ordinary normalised log-volumes of the lattices involved are computed here; what is not computed is the hull of the union of the possible images of the actual Θ-pilot, and what is not compared is the two sides of [IUT3, Cor. 3.12]. The relation of this datum to the estimates, if any, is left to a further report; *the construction stands whatever that relation turns out to be.*

20 Reproduction

`scripts/imq_checks.py` reproduces the exact arithmetic of §3 (the six orbit comparisons, the exact j, the orders at π and $\bar{\pi}$, the norms and the Frobenius traces) using only the standard library. *It does not reproduce the image argument*: the closure computation it also prints is an auxiliary check on a chosen companion representative, labelled as such in its output, and the basis-free argument is `lean/BasisFreeImage.lean`. `scripts/imq_family.py` checks the three facts used in Proposition 2 and the excluded density. `scripts/tables.py` prints every table of §8–§17, with the normalisations asserted rather than printed, and exits zero; the fourteen levels of §13 and the sweep of §9 are its Tables 11 and 12, each checking its own conclusion by `assert` — that the window holds exactly fourteen primes, that the two upper constraints are simultaneously tight at 157, and that at the two curves reaching the threshold in the tabulated range the place must be put into I^*. *The counts of the sweep itself are transcribed, not recomputed there.* `scripts/imq_sieve7.cpp` performs the seventh-power-free sieve *for the cofactor C of the datum of* (1) *over* $\mathbb{Q}(i)$, not for [First]. It is self-contained — the only operation it needs on C is division by a machine word, done on the decimal representation — and runs in about a second.

`lean/BasisFreeImage.lean` carries the basis-free image argument of §3: twelve theorems, six of which depend on no axiom at all and six only on `propext`, with no `sorryAx` and no `Classical.choice`.

Independently of those programs, two files under `lean/` — `ImaginaryQuadraticDatum.lean` and `ModSevenImage.lean` — re-prove in the Lean 4 kernel, importing nothing beyond `Init`, the integer identities and the finite predicates of §3: the norm factorisations, the valuations, the point counts and traces a_q at the witness places (recomputed there from the curve rather than read off from this report), and the finite group computations. *The residue degrees are not among them*; they are computed in `scripts/tables.py`, Table 13.

[P] *One doc-string in those files is out of date, deliberately.* Seven of the fourteen are pinned by SHA-256 to the bytes released with [First], and the auditor treats any difference as an error, so a comment inside them cannot be edited without giving up that guarantee. `ModSevenImage.lean` accordingly still carries two descriptions we have since retracted: its header presents the companion-matrix closure as an assertion about the mod-7 image, and a doc-string says that the two places above a split q "give different traces". Both are corrected here and neither enters a proof — the file's theorems are finite computations about a chosen representative, and the traces are computed there from the curve. The image argument is `BasisFreeImage.lean`, whose own header states that it replaces the earlier one; the counterexample to the trace claim, at $q = 233$, is in §3. *Four things are deliberately not in the kernel*, and the trust chain is only as short as this list allows: the seventh-power-free sieve is not run there (what is proved is that $34{,}869{,}670 = \lfloor C^{1/7} \rfloor$, which is what makes the external search exhaustive); the identification of these numeric predicates with the actual Galois representation, places and cusps of the named curve is a field of the interface below, not a theorem; the geometric arguments of §4 are not verified by these programs; and *Proposition 2 is formalised only in part. That last one is worth stating plainly.* Its proof combines finite algebra with Hensel lifting, a sieve over all primes, a density bound and an asymptotic; *only the finite algebra is in the kernel.* `lean/FamilyArithmetic.lean` carries three theorems — that a prime dividing both r^2+1 and $(r-1)^2+1$ divides 5, its contrapositive, and the factorisation $400 = 2^4 \cdot 5^2$ — with no `sorryAx` and no `Classical.choice`. *The third is a numeric identity and not a discriminant theorem*: neither f nor the discriminant occurs in its statement, and $\mathrm{disc}(f) = 400$ is computed by ordinary means outside Lean. *The rest is prose.* The density bound and the infinitude of surviving x are *not formalised here* — which is a statement about this development, not a claim that they could not be. That matters, because a gap in the prose part passes every machine check in this bundle — as one did: the root count needed $q \nmid M$, the case $q = p$ was missing, and a later review, not the kernel, found it. *The theorem now in the kernel is exactly the one that was missing.* Four further files treat §4 in the manner of an abstract interface: the cited results of anabelian geometry are the fields of a structure `AnabelianInput`, each carrying its locator, and the corehood of the datum's orbicurve is derived relative to any structure satisfying it. Three things there are proved outright, with no interface and no assumption: that $j(E_0)$ is irrational, by an exact computation in which the unreduced numerator of the imaginary part of $j(E_0)$ has 298 digits (the reduced fraction reported in §3 has 296 and 234) — so that the antecedent of [MFHMP, Prop. 2.1] is *discharged* rather than assumed, the four exceptional invariants being rational; that $\mathrm{SL}_2(\mathbb{F}_7)$ acts transitively on the 24 marked flags, which is what lets one global quotient and cusp be matched at every bad place at once; and that $2\xi \neq \pm\xi$ for every nonzero ξ, which separates the cusp 2ϵ from ϵ. The file also records two checks: a model in which the interface holds non-vacuously, and a model showing that dropping [MFHMP, Prop. 2.1] makes the conclusion underivable.

Conditions (e) and (f) are derived in the same style. The section $\mathbb{V} \to \mathbb{V}_{\mathrm{mod}}$ is proved *bijective* rather than assumed, from the transitivity above; the local conditions at the bad places and the arrow-marked covers follow from seven further interface fields, each carrying its locator in [IUT1] or [EtTh]. Each of the seven comes with two checks: a model in which its premise is actually inhabited, and a model in which that one field is deleted, the other six still hold, and the conclusion fails.

[P] *Two kinds of arithmetic occur in these scripts and we do not call both exact.* The valuations, norms, residue degrees, point counts, ramification indices, tuple coefficients and the right wall $N/3$ are integer or `Fraction` computations and are exact. The heights, the logarithms in condition (3) and the ratio of §17 are IEEE double evaluations of log and $\sqrt{\ }$, and are exact only to display precision. *Where a floating-point value decides a comparison, the margin is wide*: the left wall gives $\sqrt{H_\forall} = 85.58\ldots$ against the smallest admissible $\ell = 89$, and condition (3) holds by a factor near 7.9 across the window. *Those margins are for the fourteen-level window only*; elsewhere they are smaller — at $N = 114$, $\ell = 37$ the gap to the left wall is $2.10\ldots$ and condition (3) holds by a factor of $1.31\ldots$ — so we do not claim a single margin for the report. No verdict here is claimed to follow from an interval certificate, which we have not computed.

`scripts/lean_audit.py` checks *every* declaration with `#print axioms`, including those the elaborator generates. Its default scope is all *fourteen* files: the run of 2026-09-07 reports 263 stated theorems and 768 declarations in all, of which 565 depend on no axiom, 138 on `propext` and 65 on `propext` and `Quot.sound`, and returns PASS. Of those, `BasisFreeImage.lean` contributes 12 stated theorems and 37 declarations, the twelve splitting 6 axiom-free and 6 on `propext`. *An earlier version of the auditor named only eleven files and so did not cover that one*, which is the file carrying the argument of §3; the omission was found in a later review and is corrected. Seven of the fourteen — the files of the first report — are additionally pinned by SHA-256, and carry 140 stated theorems and 481 declarations; their axiom split 327/107/47 is what the auditor prints under `--through GeometricModels`, which it labels a partial scope. Nothing depends on `sorryAx` or on `Classical.choice`; those two are not in the auditor's allowed set, so their appearance is a failure rather than a note. *What the Lean files do not cover is stated in their headers*: the cited anabelian results are assumed and not proved; the links between the abstract interface and the concrete curve — the j of §3, and the finite predicates standing for places, flags and cusps — are named fields of the structures rather than theorems; and the sieve is not run, what is proved being that $34869670 = \lfloor C^{1/7} \rfloor$, which is what makes the search exhaustive. The geometric arguments of §4 are not verified by these programs.

Remark 8 (related work)**.** [S] The LANA project reports that its plan is to work first toward the formalization of the basic parts of anabelian geometry, and that its July 2026 interim report provides a basis for eventual formalization in Lean rather than carrying one out. [P] The Lean development described above does something different and independent: it re-proves in the kernel the arithmetic of two explicit data against [IUT1, Def. 3.1], and states the results of anabelian geometry it uses as the fields of an interface rather than proving them. Were every field of that interface proved, the conditional statements here would become unconditional. We do not claim that those fields are what LANA is formalizing: besides the anabelian theorems cited, the interface also carries the identification of the abstract places, labels and cusps with those of a named curve (§20), which is a separate obligation.

Disclosure

Large language models were used substantially in locating passages, cross-checking computations, drafting, and adversarial review; the author is responsible for the claims. The arithmetic of §3 and §5 was verified by the author's programs; the geometric arguments of §4 were drafted with machine assistance against the sources cited there and then checked against those sources. The correction history of [First] applies here as well, since the present datum reuses its structure. Several statements in earlier drafts of the tabulations of §8–§17 were withdrawn after external review: an incorrect ramification index, an omitted hypothesis of [IUT1, Def. 3.1(c)], a table applying a formula outside its domain, and a sieve analysis reduced by a cyclotomic factorisation to a finite blockwise sieve. The corrections are marked where they occur, except that the caveat on the four $\mathbb{Q}$-rows of §9 stood only in the script until this version.

References

[First] T. Urahigashi, *An explicit initial Θ-datum with mixed reduction at a split prime*, version 0.1.3-draft (2026), DOI 10.5281/zenodo.22537342 (all versions).

[CanLift] S. Mochizuki, *The absolute anabelian geometry of canonical curves*, Doc. Math., Kazuya Kato's fiftieth birthday, Extra Vol. (2003), 609–640.

[EtTh] S. Mochizuki, *The étale theta function and its Frobenioid-theoretic manifestations*, Publ. Res. Inst. Math. Sci. **45** (2009), no. 1, 227–349.

[IUT1] S. Mochizuki, *Inter-universal Teichmüller theory I*, Publ. Res. Inst. Math. Sci. **57** (2021), 3–207.

[IUT2] S. Mochizuki, *Inter-universal Teichmüller theory II*, Publ. Res. Inst. Math. Sci. **57** (2021), 209–401.

[IUT3] S. Mochizuki, *Inter-universal Teichmüller theory III*, Publ. Res. Inst. Math. Sci. **57** (2021), 403–626.

[IUT4] S. Mochizuki, *Inter-universal Teichmüller theory IV*, Publ. Res. Inst. Math. Sci. **57** (2021), 627–723.

[LANA] Project LANA, *Project LANA interim report on IUT theory*, July 2026, 50 pp.

[Rpt] S. Mochizuki, *Report on discussions, held during the period March 15–20, 2018, concerning inter-universal Teichmüller theory*, RIMS, Kyoto University, February 2019 version, 45 pp.

[SS] P. Scholze, J. Stix, *Why abc is still a conjecture*, manuscript, July 16, 2018.

[DH25] T. Dupuy, A. Hilado, *The statement of Mochizuki's Corollary 3.12: initial theta data*, arXiv:2004.13228v2 [math.NT], 19 June 2025, 20 pp. (Version 1, 28 April 2020, is a different and longer paper.)

[DHPS] T. Dupuy, A. Hilado, *Probabilistic Szpiro, baby Szpiro, and explicit Szpiro from Mochizuki's Corollary 3.12*, arXiv:2004.13108v2 [math.NT], 29 April 2020.

[MFHMP] S. Mochizuki, I. Fesenko, Y. Hoshi, A. Minamide, W. Porowski, *Explicit estimates in inter-universal Teichmüller theory*, Kodai Math. J. **45** (2022), no. 2, 175–236, DOI 10.2996/kmj45201.